



HK IMO Training 2023–2024
Problem Set 18
For 24 February 2024



Problem 69. Let p be a prime number. Define a sequence $\{a_n\}$ by $a_1 = p + 1$ and

$$a_{n+1} = a_1 a_2 \cdots a_n + p$$

for $n \geq 1$. Find all p such that the sequence contains at least one perfect square.

Problem 70. In a 1000×1000 board, each cell is coloured black or white. It is known that there exists a 10×10 sub-board such that every cell is black, and there exists a 10×10 sub-board such that every cell is white. For each 10×10 sub-board, we define its *score* to be the difference of the number of black cells and the number of white cells in this sub-board. Let N be the minimum score among all 10×10 sub-boards. Find the maximum possible value of N .

Problem 71. In $\triangle ABC$, let D be the foot of altitude from A . Let E and F be two points on the line AB such that $BD = BE = BF$, where E lies between A and B . Let I and J be the incentre and A -excentre of $\triangle ABC$ respectively. Prove that there exist two points P and Q on the circumcircle of $\triangle ABC$ such that $PB = QC$ and $\triangle PEI \sim \triangle QFJ$.

Problem 72. Let $n \geq 2$ be an even positive integer, and let $x_1, x_2, \dots, x_n$ be nonnegative real numbers with sum 1. Find the maximum possible value of

$$\sum_{1 \leq i < j \leq n} \min \{ (i - j)^2, (n + i - j)^2 \} x_i x_j.$$